

Research article

Towards decarbonizing large-scale industries: A decision support framework for optimizing grid-connected PV-battery energy systems planning – Case study of an OCP mining site, Morocco, North Africa

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ABSTRACT

Incorporating renewable energy into forthcoming grid-connected or decentralized energy systems assumes an escalating significance and can potentially enhance endeavors toward accomplishing Sustainable Development Goal 7 (SDG7). Nevertheless, deploying sustainable renewable energy-intensive systems may present challenges related to their intermittency and cost instability. This paper proposes the development of a decision support model aimed at planning an optimal on-grid PV battery system. The model builds upon the Open Energy Modeling Framework (OEMOF) and defines asset capacities, energy dispatch, operational strategies, and optimal costs for the designated system. Key factors considered include energy demand profile, photovoltaic potential, electricity tariffs, and the availability of the National Grid. Furthermore, the model evaluation incorporates the computation of pertinent performance, feasibility, and viability indicators, notably the Levelized Cost of Energy (LCOE). To validate the framework, the article conducts a case study to determine the optimal sizing and planning of a grid-connected PV battery energy system. The objective is to cater to the electricity needs of an OCP (Office Chérifien des Phosphates) mining site in Morocco. The study considers the characteristics of the national power grid, incorporating time-varying electricity tariffs based on a Demand-Response program taking into account various rates according to the time of use (ToU). The case study includes a sensitivity analysis that examines different factors, namely the proportion of renewable energy, the investment costs of photovoltaic systems and batteries, and the financial parameter known as the weighted average cost of capital (WACC). Through this analysis, the study assesses the impact of these variables on the calculated cost of energy. Experimental results revealed a range of LCOE values from \$0.07111/kWh to \$0.11847/kWh for high renewable energies integration.

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Nomenclatures

AC	alternative current	NG	national grid
BESS	battery energy storage system	NPC	net present cost
BG	biomass generator	NPV	net present value
BIPV	building integrated photovoltaic	OCF	office chérifien des phosphates
BSA	backtracking search algorithm	OEMOF	open energy modeling framework
CAPEX	capital expenditure	ONEE	office national de l'électricité et de l'eau potable
COE	cost of energy	OPEX	operational expenditure
CO ₂	carbon dioxide	PESA	pareto envelope-based selection algorithm
CPV	concentrated photovoltaic	PSO	particle swarm optimization
CRF	capital recovery factor	PV	photovoltaic
CSP	concentrated solar power	RE	renewable energy
DC	direct current	REF	renewable energy factor
DG	diesel generator	RES	renewable energy sources
DHI	diffuse horizontal irradiance	RS	renewable share
DLP	deterministic linear programming	SAM	system advisor model
DNI	direct normal irradiance	SDGS	sustainable development goals
DR	demand response	SOC	state of charge
EO	equilibrium optimizer	SST	solar steam turbine
EVCS	electric vehicle charging system	TAC	total annualized cost
GA	genetic algorithm	TMY	typical meteorological
GHG	greenhouse gas	SOO	Single objective optimization
GHI	global horizontal irradiance	SP	stochastic programming
HMG	hybrid microgrid	ToU	time of use
IWO	invasive weed optimization	UN	united nations
LARO	leader artificial rabbits optimization	WACC	weighted average cost of capital
LCC	life cycle cost	WT	wind turbine
LCOE	levelized cost of energy	x	system's component
LP	linear programming	t	1 h time step
LPSP	loss of power supply probability	a _x	asset annuity
MCDM	multiple criteria decision making	CAP _x	asset capacity
MGs	microgrids	C _{var,x}	dispatch cost
MOO	multi-objective optimization	E _x	asset dispatch
MOPSO	multiple objective particle swarm optimization	T	project duration
MINLP	mixed integer nonlinear programming	C _{res}	residual value
MILP	mixed integer linear programming	t _x	asset lifetime
MIP	mixed integer programming	d	discount factor
MIQCP	mixed integer quadratically constrained programming	E _x	asset generation value
		E _{batt,Dis}	battery discharge
		E _{batt, Char}	battery charge
		η	conversion efficiency
		E _{short}	energy flow to balance out supply shortage
		E _{ex}	energy flow to balance out excess generation
		E _{Dem}	energy demand

1. Introduction

As we reach the halfway point to the agenda 2030 deadline, a review of advances on the Sustainable Development Goals (SDGs) reveals significant challenges. According to the latest SDG progress report [1], more than half of the SDG targets show insufficient headway, while almost 30 % of improvements have stagnated or regressed. These worrying figures particularly concern the targets of reducing poverty, hunger, and global warming. This analysis highlights the urgency of stepping up efforts to ensure that each of the SDGs stays on course and moves towards a sustainable future for all.

Addressing today's key environmental challenges, SDG 7 aims to guarantee that modern society functions in harmony and develops equitably. We need affordable and reliable energy services without depending on inexpensive but energy-intensive fossil fuels, the main producers of GHG emissions that cause climate change [2]. It is therefore essential to redouble our efforts so as not to lose sight of the 2030 target and move towards a just energy transition.

As part of the energy transition, the mining industry contributes to the UN SDGs. This is by facilitating large-scale economic development and providing minerals for technology, infrastructure, energy, and agriculture. In this context, it is imperative to minimize waste production, conduct responsible mining operations, implement safer procedures, integrate sustainable technologies, reduce emissions, and perform effective environmental management [3]. As a result, the mining industry can help address many of the issues raised by the SDGs, notably environmental degradation.

Companies can actively support SDGs by integrating them into their long-term operations and fundamental plans. According to Ref. [4], only a small minority of mining enterprises report on their progress against the indicators or goals they have set. SDG7 stands as a target where the global mining industry still has a long way to go to put in place strategic measures to help achieve it.

Several research studies, including Tiainen et al. [5], Shen et al. [6], Tiainen et al. [7], and Govindan et al. [8], have investigated the impact of the mining factories toward greener production, social responsibility standards, and environmental performance. Norgate and Haque [9] state that over the last 20 years, the mining industry has recognized the significance of promoting sustainability

Table 1

Literature review of some case studies for energy system models relating to Morocco.

Case study	Year	Location	sector	Algorithm/Software	System configuration	Objective function	ToU
[30]	2019	Morocco, Tahla	Residential	Homer	PV/WT/DG/NG/BESS	minimizing the total NPC	No
[31]	2019	Morocco, Tetouan	Residential	Homer	PV/NG/BESS	minimizing the microgrid configuration cost	No
[32]	2019	Morocco, Laayoune region	–	hybrid Genetic Algorithm-PSO algorithm	WT/PV/CPV/SST/NG/BESS	sizing optimization	No
[33]	2020	Morocco, Essaouira	Residential	Multimodal Delayed PSO Algorithm	PV/WT/DG/BESS	minimizing the LCOE, minimizing the LPSP, maximizing the REF	No
[34]	2020	Morocco, Rabat	Residential	PSO	PV/WT/DG/BESS	Minimizing the NPC	No
[35]	2021	Iraq, Baghdad	Residential	Homer	PV/BESS/BG	-minimizing the life cycle costs	No
[36]	2021	Morocco, Rabat	–	MOPSO, PESA II, and SPEA2 algorithm	PV/WT/DG/BESS	- maximizing the annual reduction of CO2 emissions	No
[37]	2021	Morocco, Dakhla	–	EO	PV/WT/DG/BESS	minimizing the Net Present Cost (NPC), the penalty cost of emission and the quantity of the CO2	No
[38]	2021	Morocco, Dakhla	Residential	IWO/BSA Algorithm	PV/WT/BG/DG/NG/BESS	minimizing the NPC	No
[39]	2022	Morocco, Oujda	–	Electric System Cascade Extended Analysis/SAM	PV/WT/CSP/NG	minimizing the LPSP, the LCC and the LCOE.	No
[40]	2022	Morocco, Ouarzazate, Oujda	Residential	PSO	PV/WT/DG/BESS	minimum cost of energy (COE)	No
[41]	2023	Spain, Granada	–	–	–	–	–
[42]	2023	Algeria, Bechar	Residential	Homer	PV/WT/NG/BESS	minimizing the total NPC	Yes
[43]	2023	Morocco, Dakhla, Casablanca, Fez	Residential	MILP	PV/NG/ESS/EVCS	minimizing the net energy cost	Yes
[44]	2024	Morocco, Benguerir, Rabat	–	–	–	–	–
[45]	2024	France, Brest	–	–	–	–	–
[46]	2024	Morocco, Ouaneskra	Rural site	Homer	PV/WT/BESS	minimizing the NPC	No
[47]	2024	Morocco, Rabat, Tangier, Fez, Ifrane, Benguerir, Ouarzazate	Residential	PSO	PV/BIPV/BESS	minimizing the TAC	No
[48]	2024	Morocco, Guelmim	Residential	LARO	PV/WT/DG/BESS	minimizing the NPC	No
[49]	2024	Morocco, Berkane	Commercial platform	Homer	PV/biogas/BESS	minimizing the NPC	Yes
-	2024 (This study)	Morocco	Mining	Oemof (MILP)	PV/BESS/NG	Minimizing the COE	Yes

and managing the environment effectively. To this end, Muduli et al. [10] assert that we can put many factors in place, such compliance with legal requirements, appealing to financial institutions, and enhancing eco-efficiency.

Therefore, effective integration of SDG 7 requires support for the early implementation of renewable energy to achieve the goal of a low-carbon economy. In addition, boosting the direct production of renewable electricity by these companies will reduce their dependence on grid power supply, eliminating the need for increased generating capacity and improving system flexibility. Furthermore, the resilience of renewable energy sources (RES) to global crises, such as the COVID-19 pandemic, further underlines their importance. Studies have shown that the renewable energy generation sector is less affected by these disruptions than conventional energy sources [11].

Besides, many articles take sustainability considerations into account when developing mining sites. However, Tsisilile Igogo et al. [12] review the literature to analyze the challenges, possibilities, and enabling strategies for renewable technology integration in mining operations. Anshul Paneri et al. [13] examine the techno-economic viability of RES for switch-on, switch-off mining, a new idea

for mobile small-scale mining, in 2021. Eduardo Vyhmeister et al. [14] evaluate the use of photovoltaic and revolutionary wind technologies to supply as much as 10 % current and projected electrical energy needs of the mining industry in Chile. A. Strazzabosco et al. [15] assess the current adoption of clean energy among the mining houses in Australia. Also, literature highlights the integration of renewable energy in the African region. Fabien Muhirwa et al. [16] reviewed research published during the period 2000 to 2022 on substantial extractive industry investment, the associated issues and sustainable development potential of rural communities residing in African mine sites.

In Morocco, renewable energies play a pivotal role in the energy strategy. The country possesses significant renewable potential, letting it meet a substantial proportion of its growing energy needs by replacing fossil fuels. Therefore, Morocco aims to increase the share of renewable energies in installed electricity capacity to over 52 % by 2030 [17]. Morocco will decrease its dependency on fossil fuel sources. The share of coal in installed capacity will drop from 38.8 % in 2020 to 22 % in 2030, while the share of oil-based thermal capacity is reduced to 9.2 % in 2030 from 16.2 % in 2020 [18].

In line with a sustainable growth strategy in the mining sector, the OCP group has implemented a responsible and innovative energy program to reduce its carbon footprint. Thanks to wind, solar, and cogeneration production, OCP has been able to cover 86 % of its needs with green energy, with a vision to reach 100 % coverage by 2030. OCP group guarantees 25 % of Morocco's clean electricity and will make a major contribution to Morocco's goal of producing 52 % of its electricity from renewable and clean sources by 2030 [19]. OCP is preparing for the next stage of energy transformation by creating industry strategies, including supporting the stable integration of the highest possible shares of electricity generation from variable renewable sources.

However, moving to RE technologies at mine sites pose a number of challenges, despite the mining industry's desire to convert and its huge impact on decarbonization efforts. These include the fluctuation and variability of the electricity supply, and the difficult conditions due to the remoteness of the mining sites. At this stage, a growing trend towards electrification systems based on microgrids (MGs) is observed. MGs promote the diffusion of renewable energy sources, and solutions to overcome their intermittency, such as electrical storage. They can operate in grid-connected or off-grid mode.

Optimal microgrid technology selection is a complex subject, mainly concerned with performance in terms of environmental, social, technological, and economic considerations [20]. The optimal combination of renewable energy and energy storage technologies with local energy demand is a major challenge. Hence, it is important to consider many factors, including the high initial investment cost [21], potential capacity limitations based on the inconsistency of the energy sources used [22], infrastructure management, and social impact [23].

In the literature, researchers have developed and discussed various approaches for optimal energy planning. Büyükoçkan and Gülleryüz [24] select the optimal renewable energy technology from different alternatives as a multi-criteria decision-making problem. Lucas Schmeling et al. [25] presents an overview of relevant publications on the optimal design of various energy systems. Meanwhile, Mohammad Fattahi et al. [26] focuses on sustainable planning of a mining supply chain network using a novel cost-efficient multistage stochastic program in which strategic and tactical planning decisions are integrated. In the other hand, different studies investigate energy systems control, Alejandro Rubio et al. [27] employed an energy system dispatch optimization model including an iterative approach for generating grid constraints, which is decoupled from the linear unit commitment problem.

This is where models and software tools must be developed and implemented to provide a proper frame for the planning of the most efficient energy systems. They allow us to chain the interactions between energy resources, determine their shares, and analyze the risks associated with price evolution. As a result, we can achieve optimal capacity, dispatch, cost, and identify the most useful performance and feasibility indicators, such as LCOE, for valuing projected energy systems. The progress made in the development of these tools and the rigorous examination of their validation forms the subject of an extensive literature. Fernando Martins et al. [28] provide an updated overview of the various modeling tools available to evaluate energy systems in smart European cities. Soner Candas et al. [29] review and compare five open-source frameworks for renewable energy system modeling.

The issue of energy systems planning has attracted worldwide interest. In Morocco, several articles have been published on the subject, and numerous case studies have been conducted. Table 1 presents a systematic comparative analysis of the case studies selected from the literature relating to Morocco. Each publication is classified according to the algorithm employed, the energy system configuration, ToU and the overall objective function of the optimization. In addition, the analysis incorporates the geographical location and industrial sector associated to each case study.

The literature review reveals a significant gap in energy systems planning for the Moroccan mining sector, which has received little attention compared to the residential sector. Current literature focuses primarily on the development of household energy systems, neglecting the substantial energy demands and unique challenges of mining operations. This neglect represents a critical omission in current research. Moreover, evaluations of electricity costs from the national grid often rely on simplified billing assumptions, incompatible with Morocco's complex billing model, which includes three tariff levels (peak hours, off-peak hours, and super off-peak hours). This simplification leads to potential inaccuracies in cost estimates, highlighting the need for a more precise approach tailored to Moroccan specifics.

Furthermore, although the use of open-source planning tools is common globally, their application in Moroccan case studies remains limited. This limitation could constrain the depth and scope of research in this area. To advance our understanding of energy systems in Morocco and develop sustainable and effective energy policies, it is crucial to create tailored solutions that consider the particularities of the Moroccan mining sector and its specific billing model.

Our main contributions focus on the development and evaluation of an innovative framework for optimizing energy systems in the Moroccan context. Firstly, we have designed a decision support framework for the optimal sizing of a grid-connected PV-battery energy system, taking into account the unique electricity tariffs in the Moroccan billing model. Based on the Open Energy Modeling Framework, this model accurately assesses system capacities, energy allocation, and optimal costs for different electrification options.

To validate our model, we conducted an in-depth techno-economic study of energy technologies, including photovoltaic (PV) systems, battery energy storage systems (BESS), and converters. This study offers a comprehensive analysis of solar demand and resource profiles within the study area, providing an in-depth assessment of the technological and economic viability of these energy solutions.

We applied our framework in a real-life case study to assess its feasibility for supplying energy to a mining site in Morocco, demonstrating the practical applicability and potential benefits of our model. Finally, we carried out a comprehensive sensitivity analysis and assessment of the potential for integrating PV and BESS systems into the Moroccan energy system. This analysis includes key financial and technical parameters such as LCOE, NPV, autonomy factor, and system reliability, providing essential information for decision-making and energy planning. These contributions aim to fill the current gaps in energy system optimization research in Morocco by proposing a rigorous approach tailored to the local context, with significant potential to inform the development of a sustainable and efficient energy policy.

It's then the objective of our contributions to theory and practice. The article is organized as follows: Section 2 presents a literature review. Section 3 introduces the optimization model where the developed software and the applied framework OEMOF are implemented. Section 4 describes the case study and the simulation result. Then, Section 5 presents and discusses simulation results, and Section 6 draws conclusions and gives an outlook on future work.

2. Literature review

Planning renewable energy systems is a complex undertaking that requires multiple factors to be considered. In addition to the technical challenges of securing a stable and reliable energy supply from intermittent renewable sources due to varying environmental conditions, there are also crucial economic considerations to be taken into account. This is particularly relevant in light of the global economic downturns, which have highlighted the importance of cost effectiveness and financial viability in sustainable energy planning.

Consequently, the drive to design sustainable and cost-effective renewable energy systems is stimulating the development of existing planning frameworks and the emergence of new ones.

It is then conceivable to delve deeper into the literature to undertake a head-to-head of some existing models. The review study in Ref. [47] examine the capabilities and applications of commercial tools that have significantly advanced the planning, modeling, optimization, and performance measurement of photovoltaic systems. The open source Dispatch and Investment Evaluation Tool with Endogenous Renewables (DIETER) is used in Ref. [48]. The authors examine the role of electricity storage in systems with a high variable RES proportion in Germany. Their findings highlight that electricity storage requirements are closely linked to costs and other supply alternatives. According to Ref. [49], the micrOrgridS model was tested to size the components and energy flows of a PV-diesel-battery HMG to the Pacific Island of Lifuka. A detailed analysis of the model was carried out and compared with the results obtained using Homer R Pro 3.11 software. Moreover, the authors in Ref. [50] employed the Calliope framework to build an energy model of the Sardinia region. They validated the model by verifying energy flows between nodes to minimize system costs as the main objective function. Furthermore, the study by Ref. [51], highlights the impact of the increasing integration of renewable energies and storage units in the national electricity grids in Jordan. PLEXOS software is used to analyze this integration and minimize system operating costs. Consequently, the results strongly recommend the use of energy storage to efficiently meet the growing demand resulting from this integration. The literature on [52] utilizes EUCAD (European Unit Commitment And Dispatch) software for developing a long-term energy model incorporating energy storage. This model aims to optimize the distribution of different technologies in a fixed European electricity system. The research conducted by Ref. [53] used STeMES to develop an optimal design and operation model for a hydrogen network. However, this model considers short-term drivers and long-term planning horizons. Likewise, the literature on [54] examines the economic benefits of the main flexibility options in the energy systems of western, northern, and central Europe through BALMOREL. The research project carried out in Ref. [55] explores different scenarios for integrating renewable energies into the energy mix of the Dominican Republic using the OSeMOSYS model. According to the results, a total investment of USD 1405.00 million would be required to reduce greenhouse gas emissions by 25 % by 2030. In the context of [56], they study the potential of electricity storage in European energy systems characterized by a high presence of renewables (over 80 %). Therefore, using the REMix optimization model, they endogenously determine electricity generation capacity and distribution, including storage technologies. Researchers at [57] have used the REMod framework to model the German electricity and heat sector in a scenario dominated by renewable energies. The result demonstrates that it is technically possible to meet the entire electricity and heat demand for the building sector with 100 % renewable energy if the heat demand is reduced by at least 60 %. Additionally, the authors in Ref. [58] carried out an analysis of possible energy futures in Egypt using two different approaches, TIMES as an optimization model and the dynamic ELTAP-EGY approach. Similarly, in Ref. [59], the researchers developed and used the oemof-moea model to analyze the Italian energy system for the year 2050 to demonstrate its main characteristics. The results show that investing in flexibility options is required to minimize carbon dioxide emissions. In the literature [60], they present the COMANDO model, illustrated by a case study of a German industrial estate. The EHDO tool is used in Ref. [61], to design and optimize multi-energy systems. Two case studies are carried out, the first on an energy center in Cologne, Germany, and the second on the supply of renewable energy to an island system on Easter Island, Chile. In addition, the authors in Ref. [62], used HOGA software to analyze three proposed renewable energy systems. The results demonstrated that the optimal configurations in terms of cost-effectiveness are PV-wind hybrid systems and stand-alone PV systems.

However, it is clear from these various case studies that optimization models can address complex questions concerning the integration of renewable energies, storage systems and demand-side management. This underlines the growing significance of optimization models within the transition to more sustainable energy systems, while also identifying the challenges for effective

Table 2
Literature review on planning techniques applied on integrated energy system.

Model/ Approach	Original publication	Year	Type of program	Model features	Case study	Description	Note
DIETER	Zerrahn & Schill [63]	2017	LP	SOO	Germany [48]	Dispatch and Investment Evaluation Tool	Open source
microGridS	Sarah Brendes [49]	2018	LP/MILP	SOO	Pacific Island Lifuka, Tonga [49]	Sizing and optimization tool	Open source
Calliope	Stefan Pfenninger and Bryn Pickering [64]	2018	MILP	SOO	Sardinia region [50]	Energy system Model	Open source
PLEXOS	Drayton et al. [65]	2004	MILP/MINLP	SOO	Jordan [51]	Integrated Energy Model	Commercial
EUCAD	Jacques Després et al. [52]	2014	MIQCP	SOO	Europe [52]	Optimization model	–
STeMES	Sheila Samsatli et al. [53]	2015	MILP	MOO	Great Britain [53]	Spatio-temporal model of energy systems	–
Balmorel	Hans Ravn et al. [66]	2001	LP (MIP)	SOO	Europe [54]	Energy system model	Open source
OSeMOSYS	Mark Howells [67]	2011	LP	SOO	Dominican Republic [55]	Energy modelling system. System optimization model	open source
REMix	Scholz, Yvonne [68]	2012	DLP	SOO	Europe [56]	Renewable Energy Mix energy system model	–
ReMod	Hans-Martin Henning, Andreas Palzer [69]	2014	Mathematical–numerical optimizer	SOO	Germany [57]	Renewable Energy Model	–
TIMES	Richard Loulou & Maryse Labriet [70]	2007	SP	SOO	Egypt [58]	Energy system optimization model generator	–
oemof-moea	Matteo Giacomo Prina et al. [59]	2020	LP/MILP	MOO	Italy [59]	Multi-objective investment optimization for energy system model	–
COMANDO	Marco Langiu et al. [71]	2021	MILP	MOO	Germany [60]	Energy Systems Optimization model	open source
EHDO	Marco Wirtz et al. [61]	2020	MILP	MOO	Germany, Chile [61]	webtool for the optimal design of complex MESs	open source
HOGA	Dufo-López [72]	2004	GA	SOO	Egypt [62]	Hybrid Optimization Program	–

implementation of these approaches in practice. Table 2 details the range of different optimal planning techniques mentioned above.

3. Optimization model

3.1. Background

The optimal design and dispatching of a microgrid can be achieved using a variety of methods and at different levels, depending on the tool selected. This underlines the crucial importance of choosing the appropriate software according to the specific situation under study, the results expected, and the availability of input parameters.

Researchers have developed numerous decision-aid tools for energy system modeling and optimization. Hans-Kristian Ringkjøb et al. [73] presented a review of different energy system planning and analysis software tools. Kamal Anoune et al. [74] and Sinha.S and Chandel.SS [75] specifically evaluated planning tools as they apply to hybrid energy systems. Connolly et al. [76] investigate in detail more than 37 software tools used to examine renewable energy integration.

various tools discussed in the literature, HOMER is widely regarded as the industry standard, and extensively utilized by project designers and researchers worldwide. It's a user-friendly tool with many features such as system settings, illustrations, and output tables. Nevertheless, its primary limitation lies in its commercial and proprietary nature [77]; Programmers can't modify the tool's code. OEMOF, on the other hand, is an open-source framework for developing models of energy systems and different applications for performing energy system analysis.

3.2. Model Description

3.2.1. Open energy modelling framework

OEMOF [78] used extensively in a wide range of research questions such as [79]. This open-source tool written in the Python programming language allows for the definition of high-resolution input data and detailed analysis of the results. In particular, OEMOF enables comprehensive modeling of energy supply systems by representing them in the form of directed graphs. In these graphs, vertices correspond to system components, such as sources, sinks, and technologies, or to buses that regulate the flow of resources between these components. The edges connecting some components and buses are directed and represent the resource flows. Using

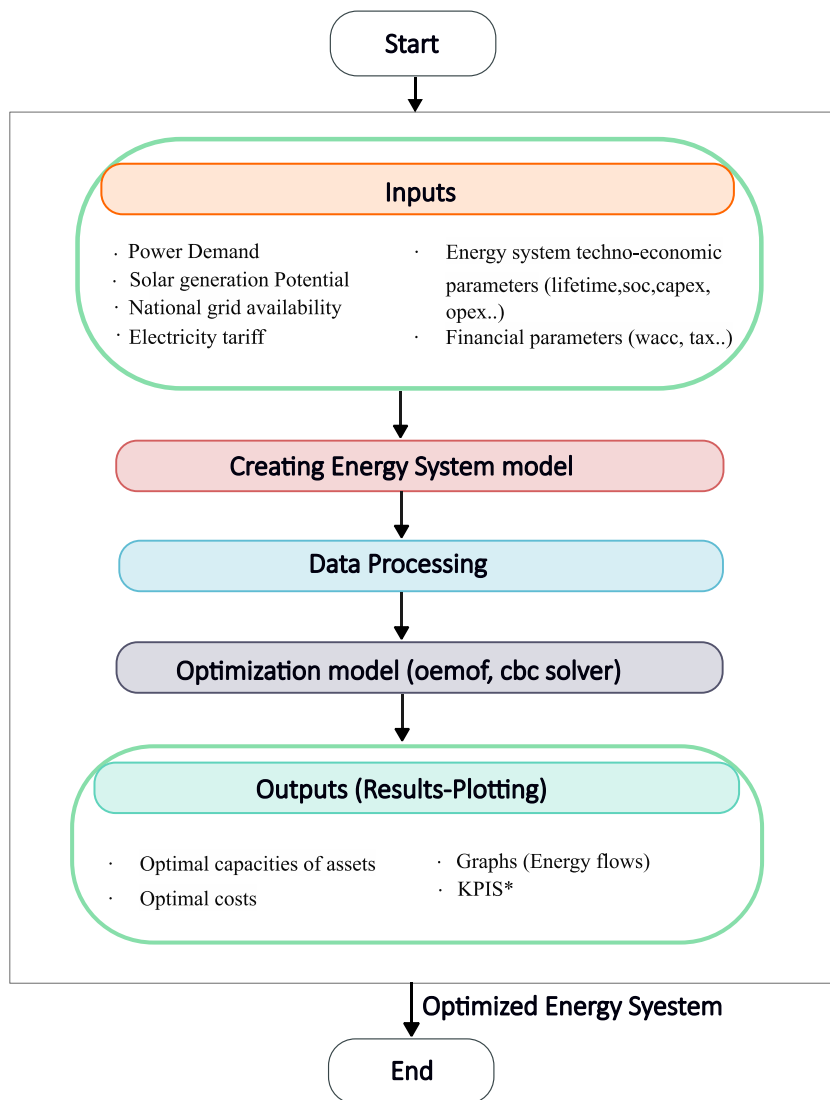


Fig. 1. The flowchart of the Framework.

*KPIS: Autonomy Factor, Supply reliability, NPV, CO2 Emission, First Investment, Total Demand Annual, Total Demand Supplied Annual, Total PV Generation, Total Charged Energy, Total Discharged Energy.

Pyomo [80], this graph is then converted into a mixed integer linear mathematical optimization problem (MILP) that can be solved numerically by a variety of solvers [81]. The goal is to satisfy all energy demands while minimizing the system costs resulting from the energy flows.

3.2.2. Proposed model

Employing the OEMOF library, we have developed an optimization framework designed to optimize capacities and flows within the energy system. This framework enables the system to meet energy demand at marginal cost-efficiently. The model's development considers the Moroccan national grid billing system for electricity tariffs, which is mainly based on consumption per time slot (peak, off-peak and on-peak) [82].

The proposed model offers a set of inputs considering.

- The technical and economic parameters relating to each system component, and the financial parameters, are maintained as fixed values.
- demand profile, generation potential, grid availability as a time series,
- And two options for electricity prices: fixed as a constant input or dynamic as a time series, allowing greater flexibility in the tool's financial model to adapt to new pricing methods.

The process of the tool is shown in Fig. 1, presenting the flowchart for the model. The main input and output elements are highlighted.

The framework determines the cost-optimal system configuration by minimizing the annual cost to supply electricity. The objective function describes this total annualized which is presented in equation (1). In the context, $\mathbf{x}$ represents all the system's components, while $\mathbf{t}$ stands for a 1-h time step over a simulation period of one year. Besides, $\mathbf{a}_x$ signifies the asset annuity per installed unit, including both investment and fixed operation and management costs. $\mathbf{CAP}_x$ refers to the asset capacity, $\mathbf{C}_{var,x}$ to the variable operational or dispatch cost, and $\mathbf{E}_x(\mathbf{t})$ to the asset dispatch.

$$\text{Min TAC} = \sum_x \mathbf{a}_x \cdot \mathbf{CAP}_x + \sum_x \sum_t \mathbf{C}_{var,x} \cdot \mathbf{E}_x(\mathbf{t}). \quad (1)$$

The annual costs specific to each system asset include replacement costs plus operating & management costs over the life of the project T and the asset's residual value at the project's end as defined in equation (2).

$$\mathbf{a}_x = \left(\mathbf{capex}_x + \sum_{n=1}^{n_{\max}} \frac{\mathbf{capex}_x}{(1+d)^{n \cdot t_x}} - \mathbf{c}_{res,x} \right) \cdot \mathbf{CRF}(T) + \mathbf{opex}_x \quad (2)$$

Equation (2) represents several parameters. Specifically, **capex** stands for investment cost, **opex** denotes the annual operation and management cost, **n** refers to the asset replacements number during the project life. Additionally, **CRF** represents the capital recovery factor, **c_{res}** denotes the residual value of an asset at the end of the project life, **t_x** stands for asset lifetime, **d** refers to the discount factor and **T** to the project duration.

The capital recovery factor (equation (3)), the number of replacements (equation (4)), and the residual value of the asset after the project ends (equation (5)) which is calculated and then deducted from the investment cost during the project, are all within the objective function.

$$\mathbf{CRF}(T) = \frac{d \cdot (1+d)^T}{(1+d)^T - 1} \quad (3)$$

$$n = \text{round} \left(\frac{T}{t_x} + 0.5 \right) - 1 \quad (4)$$

$$\mathbf{c}_{res} = \frac{\mathbf{capex}}{(1+d)^{n_{\max} \cdot t_x}} \cdot \frac{n_{\max} \cdot t_x - T}{T} \quad (5)$$

3.2.3. Microgrid constraints

Equation (6) describes the power balance constraints. The assets must be distributed in such a way as to balance the electrical buses at each time step and dispatch functions for the energy system.

$$\mathbf{E}_x(\mathbf{t}) + \mathbf{E}_{Batt,Dis}(\mathbf{t}) \cdot \eta_{Dis} - \mathbf{E}_{Batt,Char}(\mathbf{t}) + \mathbf{E}_{short}(\mathbf{t}) - \mathbf{E}_{ex}(\mathbf{t}) = \mathbf{E}_{Dem}(\mathbf{t}) \quad (6)$$

Equation (6) introduces a set of parameters within the context of the analysis. The variables are defining as follows:

$\mathbf{E}_x(\mathbf{t})$: asset generation value,

$\mathbf{E}_{Batt,Dis}(\mathbf{t})$: battery discharge after discharge losses,

$\mathbf{E}_{Batt,Char}(\mathbf{t})$: energy flow from battery charge before charge losses,

η : conversion efficiency,

$\mathbf{E}_{short}(\mathbf{t})$: reduction of energy flow in order to compensate shortage supply (Energy flow from symbolic source to balance out supply shortage),

$\mathbf{E}_{ex}(\mathbf{t})$: energy flow mitigated to balance out excess generation (Energy flow into symbolic sink to balance out excess generation),

$\mathbf{E}_{Dem}(\mathbf{t})$: energy demand.

3.2.4. Battery constraints

The battery can store energy under two conditions: The minimum and maximum storage level (equation (7)), and this energy varies as the battery is charged and discharged and as self-discharge losses occur (equation (8)).

$$\mathbf{CAP}_{Batt,KWh} \cdot \mathbf{SOC}_{\min} \leq \mathbf{E}_{Batt,stored}(\mathbf{t}) \leq \mathbf{CAP}_{Batt,KWh} \cdot \mathbf{SOC}_{\max} \forall \mathbf{t}, \quad (7)$$

$$\mathbf{E}_{Batt,stored} = \mathbf{E}_{Batt,stored}(\mathbf{t}-1) \cdot (1 - \epsilon_{sd}) + \mathbf{E}_{Batt,Char}(\mathbf{t}) \cdot \eta_{Char} - \frac{\mathbf{E}_{Batt,Dis}(\mathbf{t})}{\eta_{Dis}}, \quad (8)$$

with ϵ_{sd} self discharge losses of the battery per time step.

Equations (9) and (10) illustrate that battery charging and discharging is limited by a C-rate and the optimized charge and discharge power of the battery,

$$\mathbf{E}_{Batt,Char}(\mathbf{t}) \leq \mathbf{CAP}_{Batt,KWh} \cdot \mathbf{Crate}_{Char} \wedge \mathbf{E}_{Batt,Char} \leq \mathbf{CAP}_{Batt,KWh} \cdot \mathbf{Crate}_{Char}, \quad (9)$$

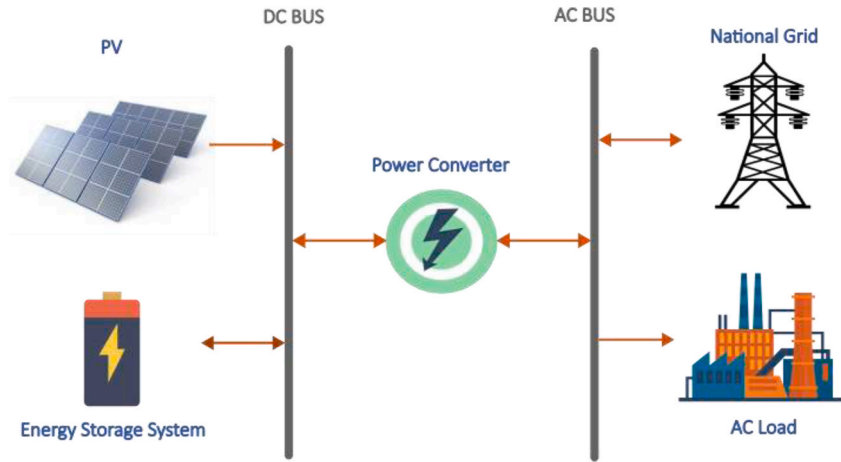


Fig. 2. The Energy system model adapted to the simulation tool.

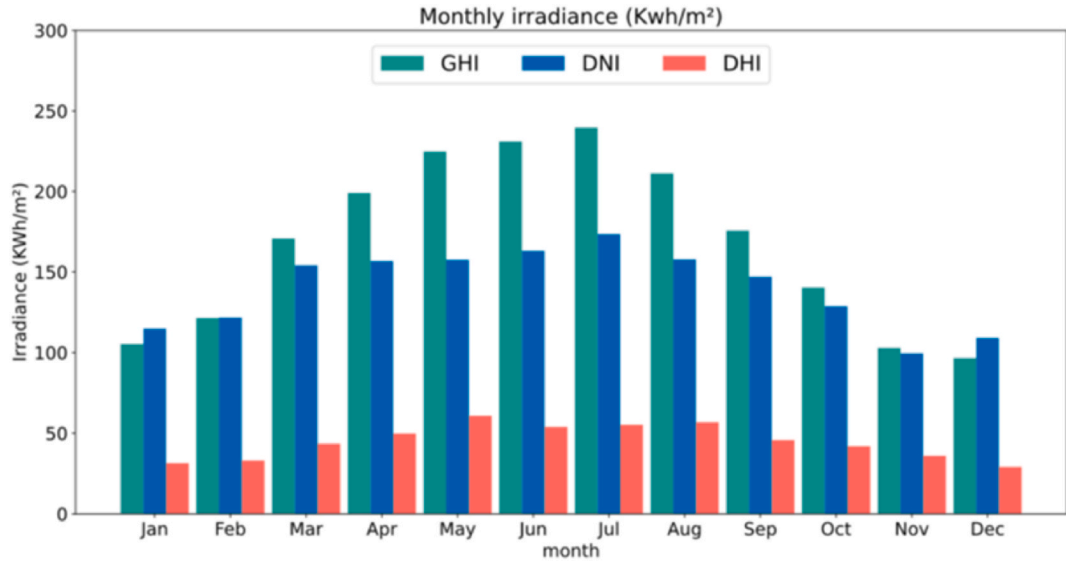


Fig. 3. Monthly global horizontal irradiance (KWh/m²).

$$E_{Batt,Dis}(t) \leq CAP_{Batt,KWh} \cdot Crate_{Dis} \wedge E_{Batt,Dis} \leq CAP_{Batt,KWh,Dis}. \quad (10)$$

The electricity stored in the battery in first- and last-time step is determined to be equal (equation (11)):

$$E_{Batt,stored}(0) = E_{Batt,stored}(T). \quad (11)$$

4. Case study

4.1. Overview

As part of the OCP strategy, we apply the modeling and optimization framework to the energy system supplying an OCP's mining site to better understand its techno-economic potential.

We define our supply option as a complete set of components (PV, battery storage system, converters, connection to the national grid). Fig. 2 illustrates the architecture of this supply system, which is evaluated based on the reliability and supply cost. Therefore, our work tackles the broader question of efficiently and optimally reducing dependence on the national grid through the implementation of a battery storage system to accommodate the intermittency of PV sources. To analyze this alternative, it is necessary to define optimal overall system capacities and dispatch strategies that can meet the electricity demand currently supplied only by the National Grid.

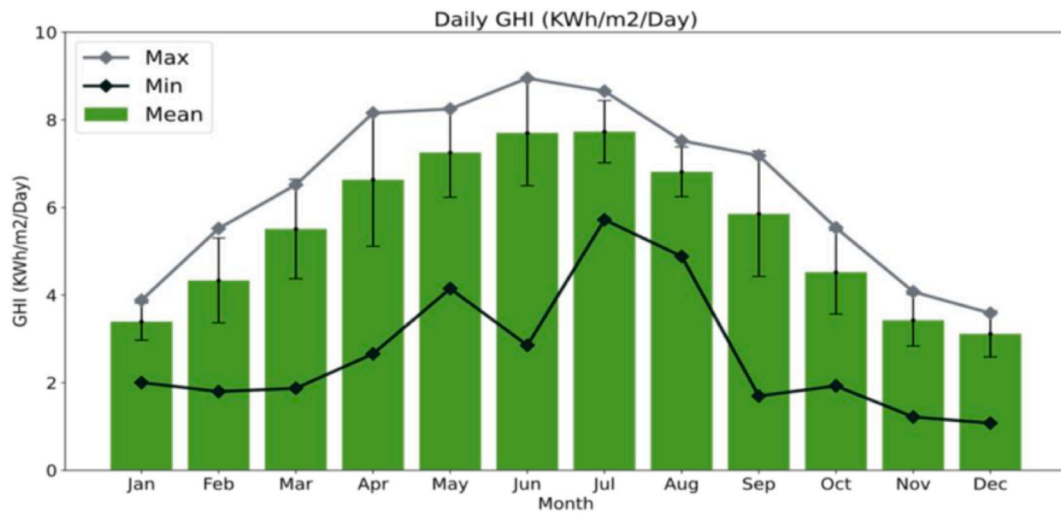


Fig. 4. Monthly average GHI values in kWh/m²/Day, min/max monthly value (lines), and variability.

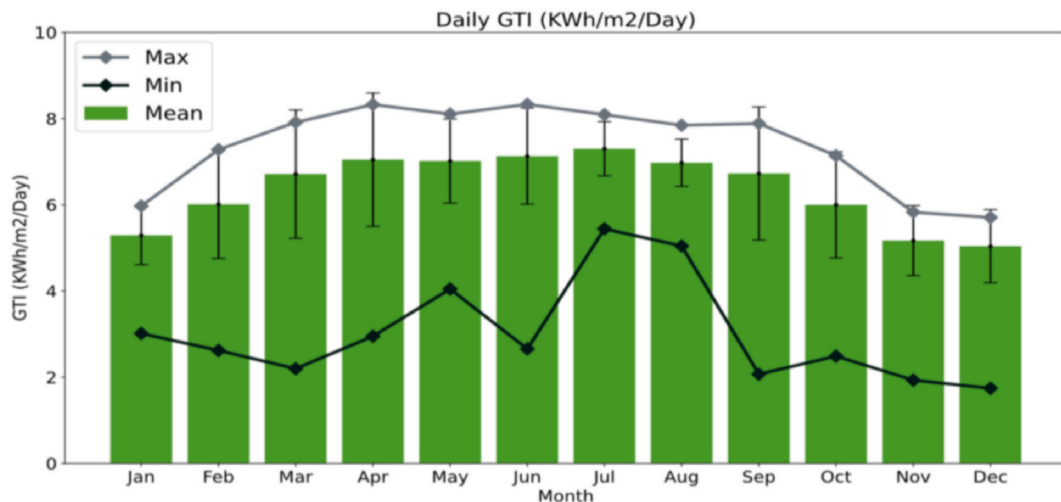


Fig. 5. Monthly average GTI values in kWh/m²/Day, min/max monthly value (lines), and variability.

4.2. Input parameters

4.2.1. Solar resources

A detailed assessment of the feasibility of the effective use of solar energy in an energy system is of primary importance. The uncertainties associated with the solar resource can make a very important distinction between a benefit and a loss, which underlines the relevance of the solar resource analysis of the chosen site. To optimally size future photovoltaic power plants, it is necessary to evaluate the solar resource using site meteorological data to simulate the yield of the solar station that will be set up at the mine site.

One of the ultimate goals of the assessment is to characterize the solar irradiance available for energy conversion for the region or location of interest over a historical period of interest. Based on the Helioclim, Helioclim3 (HC3V5) [83] database data provided in a TMY (Typical Meteorological Year) format including all geographical and meteorological parameters, a set of evaluation calculations and simulations are made.

We have used three fundamental components of solar radiation at the Earth's surface: global horizontal irradiance (GHI), direct normal irradiance (DNI), and global tilted irradiance (GTI).

Fig. 3 illustrates the calculated monthly global horizontal irradiance (GHI) (KWh/m²), we can observe that it reaches a maximum of 239.68 KWh/m²/month in July and a minimum value of 96.59 KWh/m²/month in December, the annual global irradiance of this site reaches 2.01 MWh/m².

Fig. 4 represents the variation of GHI (KWh/m²/Day) for each month. The variability around the monthly average value $\pm$ standard deviation for each month. The minimum and maximum values for each month are also shown. The site is characterized by high

Table 3
Input parameters for site Case study.

Component	Parameter	Unit	Value
PV	Cost_investment	\$/KWp	691
	Cost_opex	\$/KWp/yr	20
	Lifetime	yr	25
Storage	Cost_investment (Capacity)	\$/KWh	203
	Cost_investment (Power)	\$/KW	812
	Cost_opex	\$/KWh/yr	3.8
	Lifetime	yr	10
	Charge_efficiency	Fraction	0.98
	Discharge_efficiency	Fraction	0.98
Inverter	Cost_investment	\$/KW	174
	Cost_opex	\$/KW/yr	6
	efficiency	%	0.98
	Lifetime	yr	15

horizontal irradiance estimated at an average of 5.52 KWh/m²/Day, while it reaches a peak value of 8.95 KWh/m²/Day in June and a minimum value of 1.07 KWh/m²/Day in December. It is also observed that the most fluctuation in irradiance occurs in April and September.

Fig. 5 illustrates the variability of the global tilted irradiance (for a chosen optimal tilt angle of 30°) for each month. The average daily GTI is 6.36 KWh/m²/Day, which equals 6 full sun hours, or the operating hours of a PV system at 1KW/m² per day. The global tilted irradiance reaches a peak maximum of 8.33 KWh/m²/Day in June and min. value of 1.74 KWh/m²/Day in December.

4.2.2. Load profile

In this stage, the site energy consumption and demand profiles for the period 2019–2021 are studied quantitatively and qualitatively, with intensive analysis of annual, seasonal, monthly, weekly, and daily consumption variability. The data used in this study was collected using electricity meters provided by ONEE, the grid provider in Morocco.

The annual consumption value for 2019 appears to be lower than the relatively similar years 2020 through 2021 due to some data losses. The year 2020 was chosen as the typical year for our study.

4.2.3. System component techno-economic specifications

In the techno-economic specifications section of this study, photovoltaic (PV) and storage systems will be used as primary alternatives to grid consumption, rather than simply serving as additional backup devices. To achieve optimal energy dispatch, we will fully integrate the OEMOF framework's dispatch optimization method, which is based on perfect forecasts, into the simulation process. This integration is designed to improve the efficiency and accuracy of energy system modeling, ensuring a robust analysis of supply and demand dynamics.

The input data set includes a demand profile, grid electricity prices divided into 3-time segments (full, off-peak, and peak), a specific time series of solar production, a grid availability profile, system costs, and technical parameters (see Table 3).

We apply the projected costs and technical assumptions studied to describe the cost investment, OPEX, lifetimes, and other specified technical parameters of specific Microgrid components. For instance, we employ an initial capital cost of 691 \$/kWp, an operating cost of 3 % of the capital cost, and a 25-year lifetime for the PV plants [84]. For the storage system, we chose the lithium-ion battery technology for our study case promising lower costs, higher efficiency, and longer life cycles compared to other technologies with a CAPEX of \$203/kWh of installed capacity, OPEX of \$5/kWh/year, assigning a lifetime of 10 years [85].

4.3. Scenario development: sensitivity analyses

To conduct an economic study, we apply the analysis of multiple scenarios in the described case study. This involved performing a sensitivity analysis on some economic parameters. First, we examine the PV and storage system costs sensitivity, along with the sensitivity of renewable energy share within the system under study. Specifically, the analysis takes into account renewable energy shares of 61.18 % and 31.18 %, which have been defined by the OCP group according to its green energy requirements. These shares reflect the OCP's strategic objectives to meet its energy needs while reinforcing its commitment to sustainable development. Second, we evaluate the sensitivity of the weighted average cost of capital (WACC), a widely used financial parameter that determines the required project rate of return. By addressing these factors, we seek to gain a comprehensive understanding of the economic implications and potential outcomes associated with the studied system.

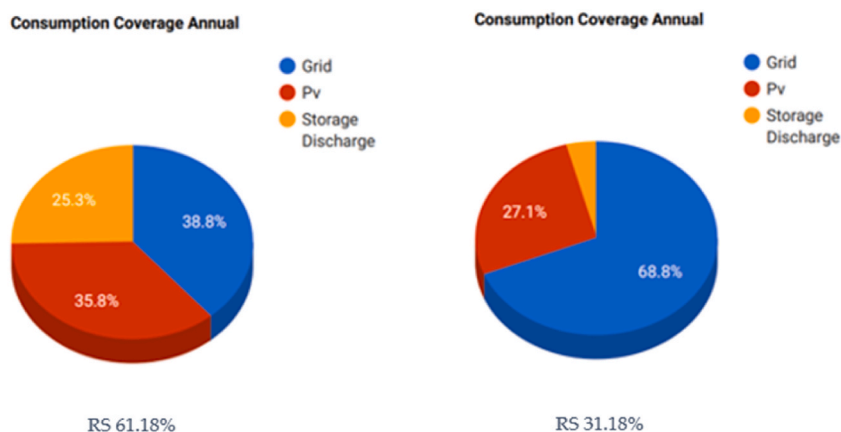
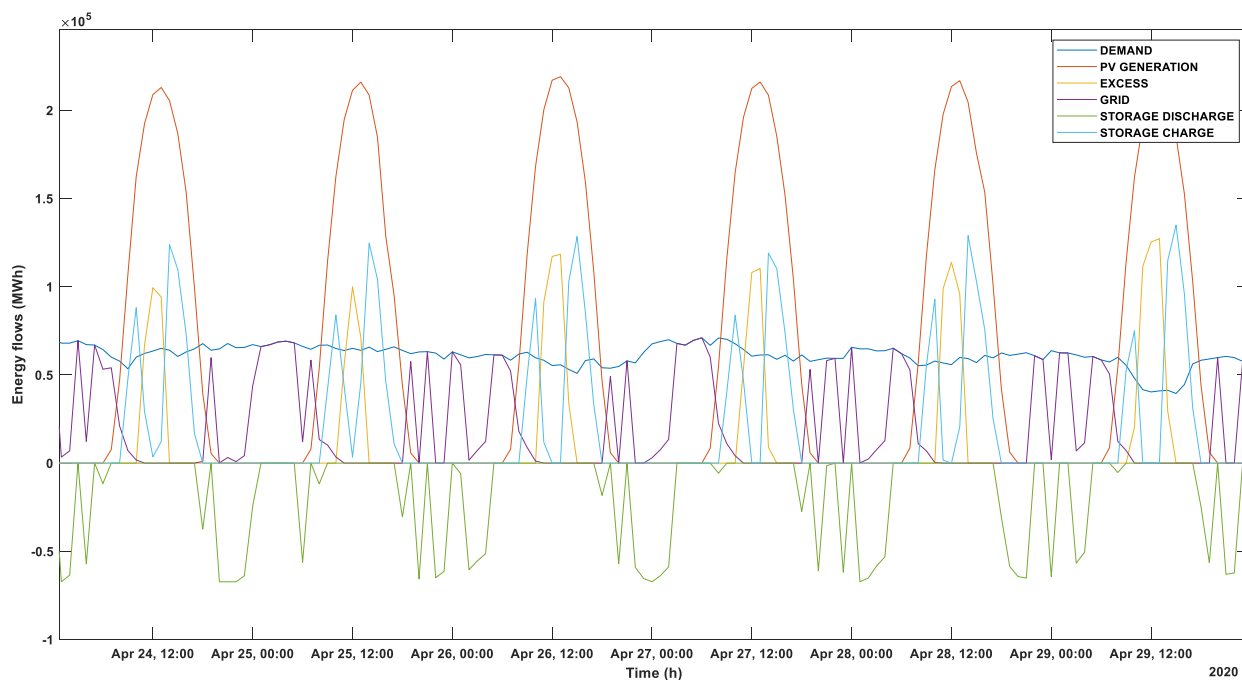
4.3.1. - Scenario 1: PV capex, renewable share (RS)

In this first scenario, we subject the PV investment cost and the share of renewables in the energy system to a sensitivity analysis. Four experiments result from the four combinations introduced by the suggested values of the PV investment cost (present value, present value-25 %) and the share of renewables (61.18 % RS, 31.18 % RS).

Table 4

Techno-economic assessment for scenario 1.

PV Capex	Present Value (691 \$/kWp)						
RS	Autonomy Factor (%)	LCOE (\$/kWh)	NPV (M\$)	PV Capacity (kWp)	Storage Capacity (kWh)	Supply Reliability (%Demand)	First investment (M\$)
61.18 %	61.18	0.11847	428.730	199.153	659.146	1	283.282
31.18 %	31.18	0.07111	257.342	100.246	68.353	1	92.838
PV Capex	Present Value – 25 % (Present Value) (518.25 \$/kWp)						
61.18 %	61.18	0.11096	401.519	212.381	635.331	1	250.895
31.18 %	31.18	0.06769	244.966	105.407	59.317	1	76.309

**Fig. 6.** Annual consumption coverage (PV capex present value).**Fig. 7.** Energy flows (PV Capex Present value, RS 61.18 %).

4.3.2. - Scenario 2: Battery Capex, RS

The second scenario concerns the investment cost of the batteries and the share of renewable energies in the studied energy system. The investment cost of the batteries takes two values (present value, present value - 25 %), and the share of renewable energies is fixed

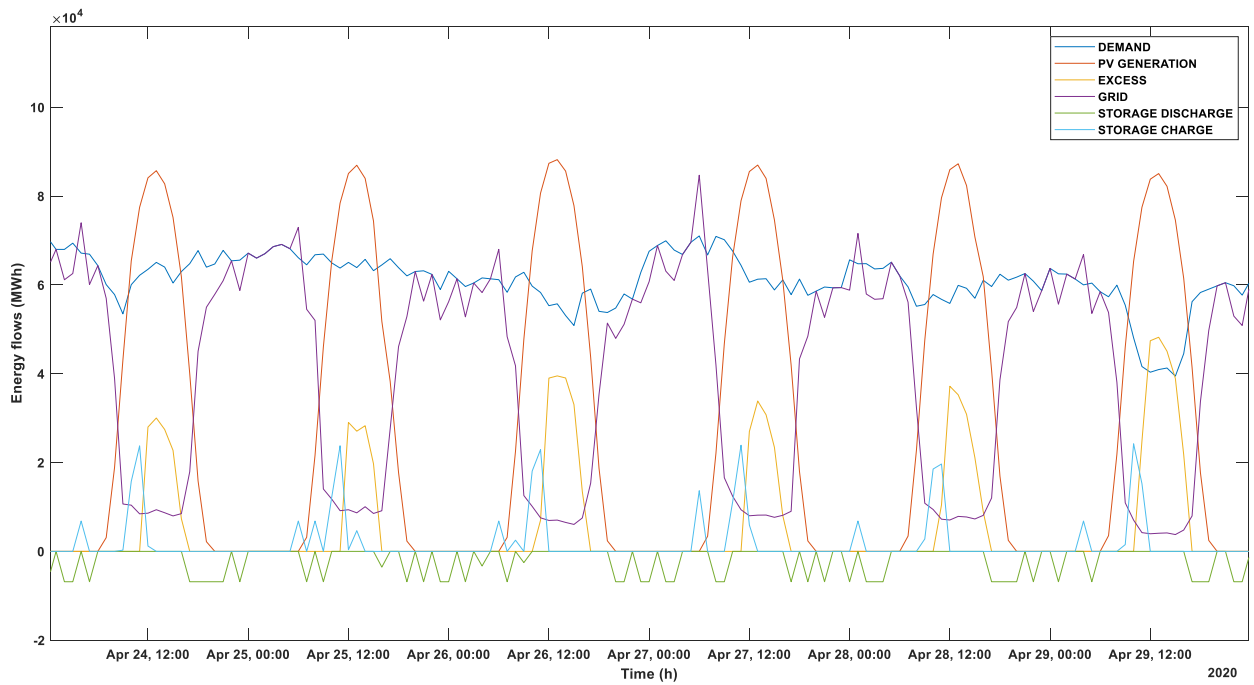


Fig. 8. Energy flows (PV Capex Present value, RS 31.18 %).

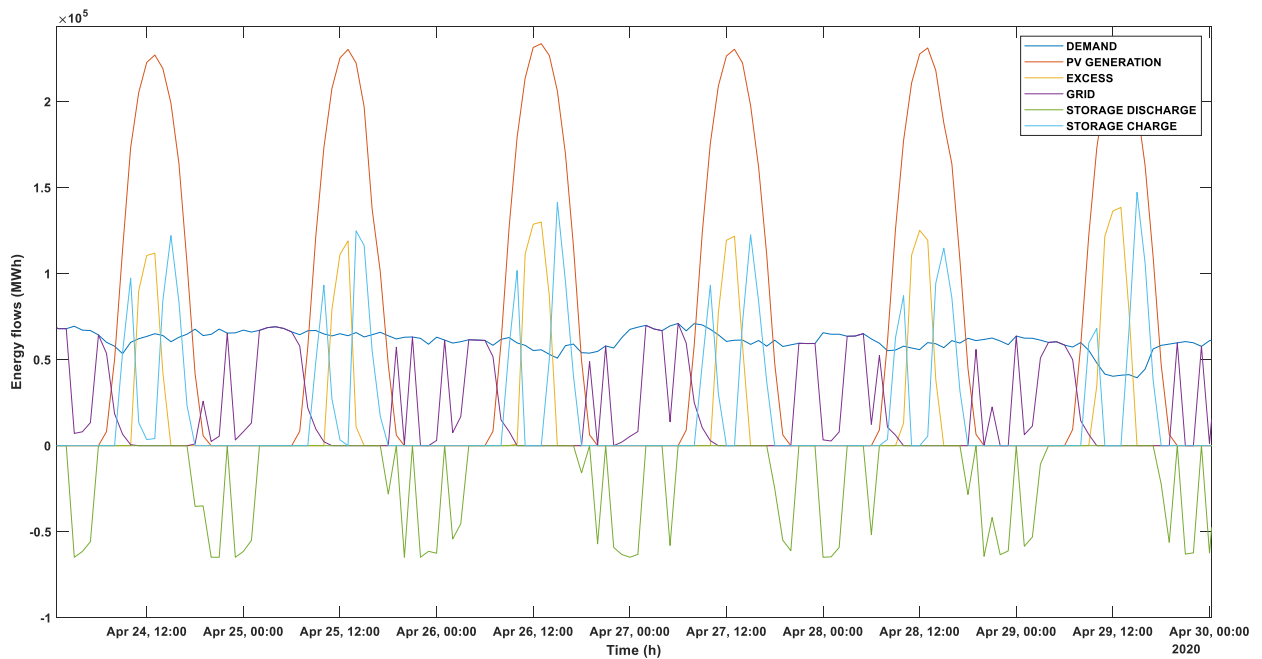


Fig. 9. Energy flows (PV Capex Present value-25 %, RS 61.18 %).

at two values (61.18 %, 31.18 %).

4.3.3. - Scenario 3: WACC, RS

The third scenario deals with the weighted average cost of capital (WACC) and the share of renewables in the studied energy system. The Wacc takes three values (5 %, 7 %), and the share of renewable energy is set to two values (61.18 %, 31.18 %). These combinations generate four sensitivity analysis experiments and thus four simulation results.

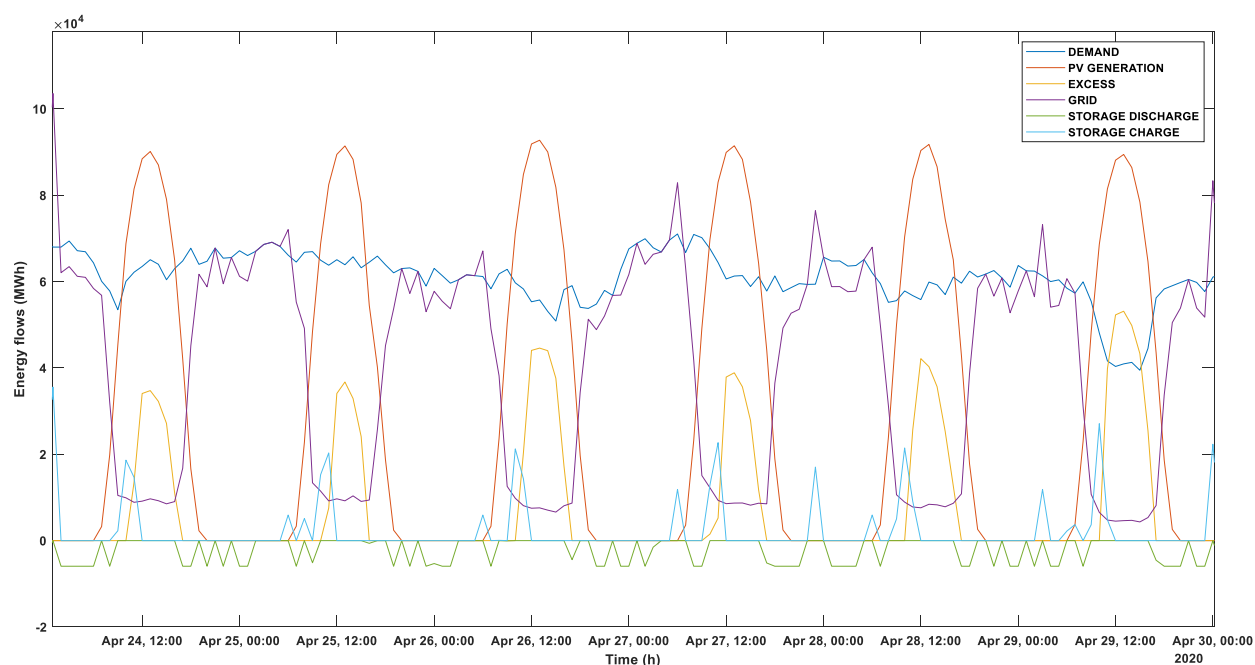


Fig. 10. Energy flows (PV Capex Present value-25 %, RS 31.18 %).

Table 5

Techno-economic assessment for scenario 2.

Battery Capex	Present Value (203 \$/kWh)						
RS	Autonomy Factor (%)	LCOE (\$/kWh)	NPV (M\$)	PV Capacity (kWp)	Storage Capacity (kWh)	Supply Reliability (% Demand)	First investment (M\$)
61.18 %	61.18	0.11847	428.730	199.153	659.146	1	283.282
31.18 %	31.18	0.07111	257.342	100.246	68.353	1	92.838
Battery Capex	Present Value – 25 % (Present Value) (152.25 \$/kWh)						
61.18 %	61.18	0.10536	381.281	190.570	678.748	1	246.750
31.18 %	31.18	0.06961	251.907	94.788	80.747	1	87.472

5. Simulation results and discussion

5.1. Scenario 1

The optimal capacities of the energy system for each sensitivity experiment are determined under scenario 1. The techno-economic optimization determines the optimal capacities and their distribution for each of the combinations (PV Capex, RS). To visualize the optimizations made for the project site, Table 4 containing all costs and optimal capacities is presented. The cost results are discussed and, where possible, compared to the results of other sensitivity experiments.

Energy demand is met directly by solar power generation and battery discharge, by a minimum of 61.18 % renewable energy (Fig. 6). The main grid is used exclusively to cover the remaining demand, which represents around 38.8 % of the total energy used on site.

Figs. 7–10 show the distribution of generation and storage components at the site under investigation. The site-specific load profile exhibits a peak demand for 83.138 MW. Based on this profile, the optimal PV system capacity varies between 100.246 and 212.381 MWp for a renewable supply of 61.18 %, while for a renewable share of 31.18 %, it ranges from 100.246 to 105.407 MWp. The PV system capacity depends on the capital cost associated with PV installation.

5.2. Scenario 2

In scenario 2, the evaluation of lithium-ion batteries cost investment is combined with the assessment of the renewable energy share within the energy system. Four sensitivity experiments were simulated. Table 5 presents the results, showcasing the cost and capacity optimizations achieved by the framework for the project site. This allows a clear view of the cost and capacity optimizations

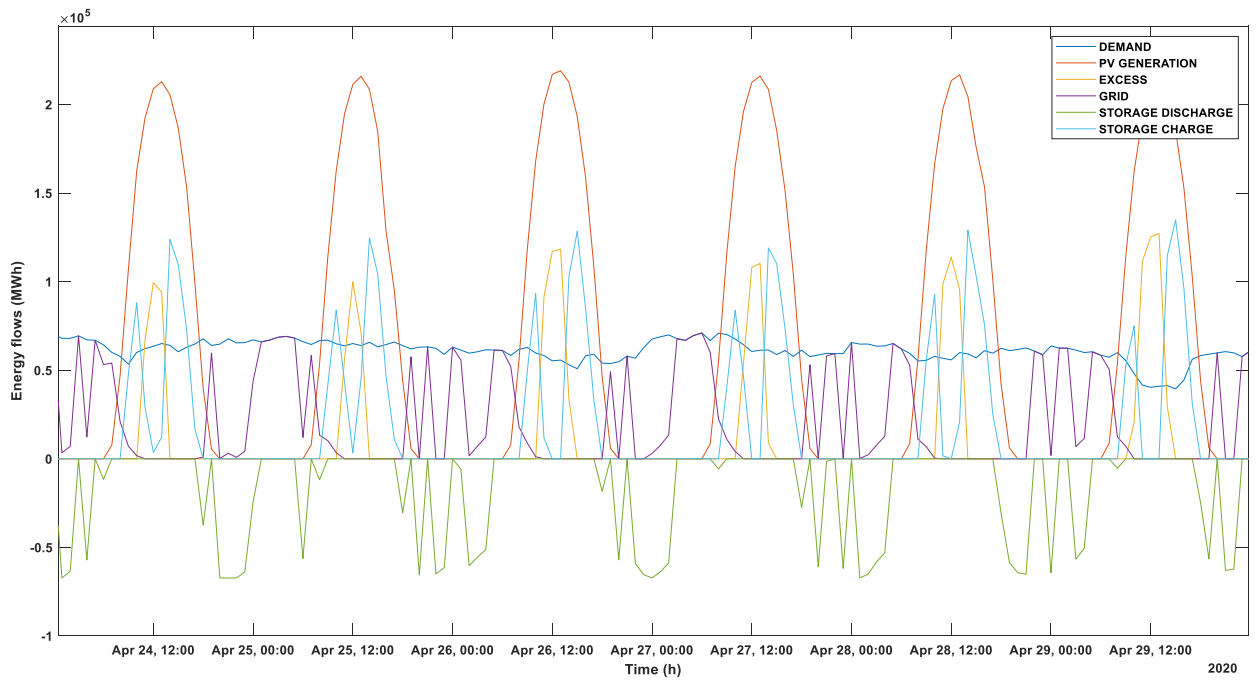


Fig. 11. Energy flows (Battery Capex Present value, RS 61.18 %).

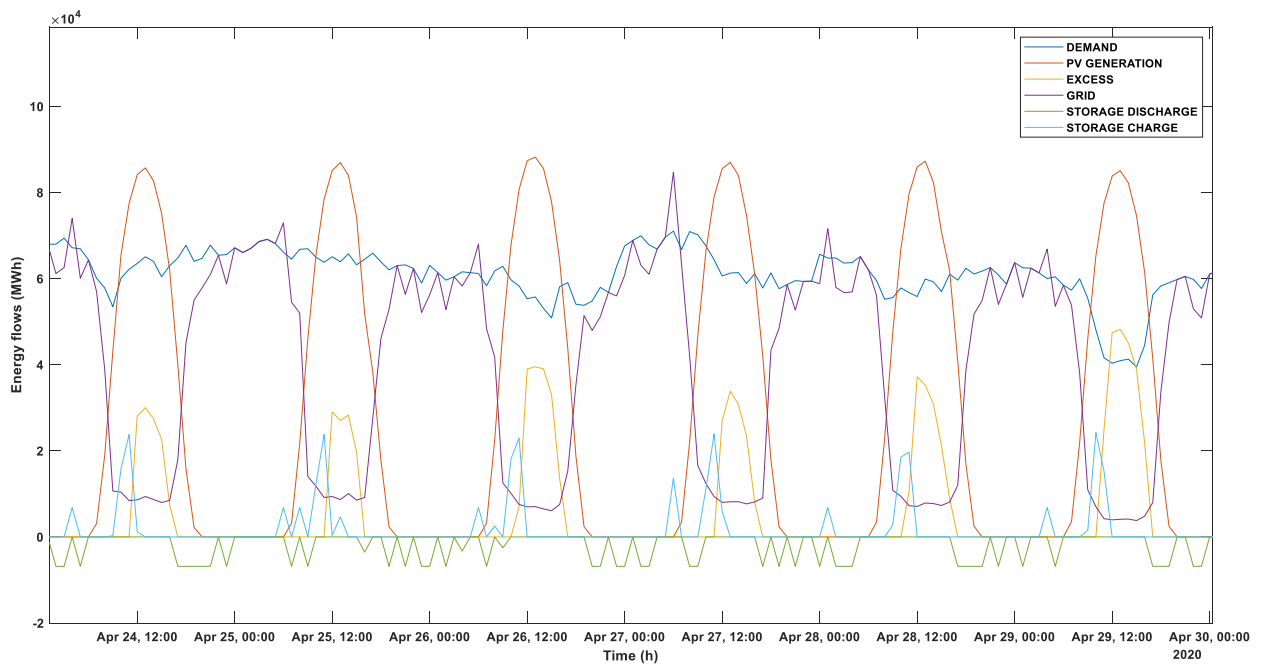


Fig. 12. Energy flows (Battery Capex Present value, RS 31.18 %).

made by oemof for the project site.

Figs. 11–14 illustrate the distribution of the generation and storage components at the site. The optimal capacity for the PV system, depending on the PV capital cost and the renewable share of supply, varies between 190.570 and 199.153 MWp for a renewable supply of 61.18 %. In the case of 31.18 % renewable supply, the optimal capacity ranges from 94.788 to 100.246 MWp. These variations are linked to the site-specific load profile, characterized by a peak demand of 83.138 MW.

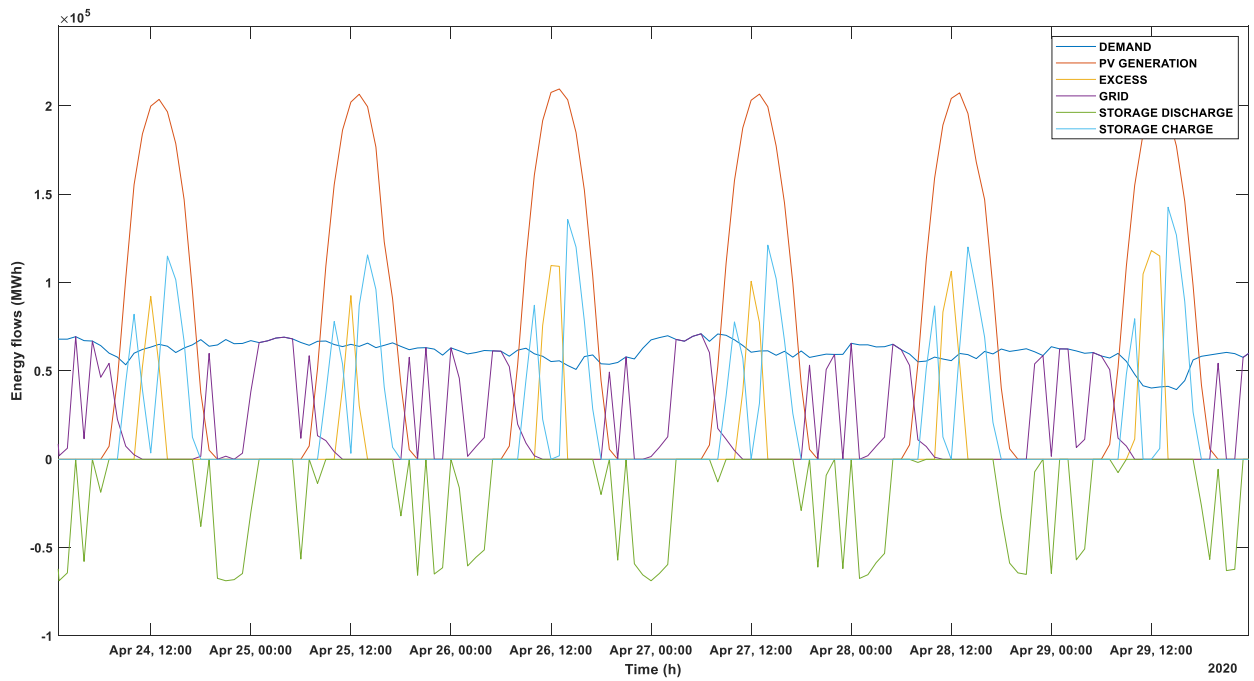


Fig. 13. Energy flows (Battery Capex Present value-25 %, RS 61.18 %).

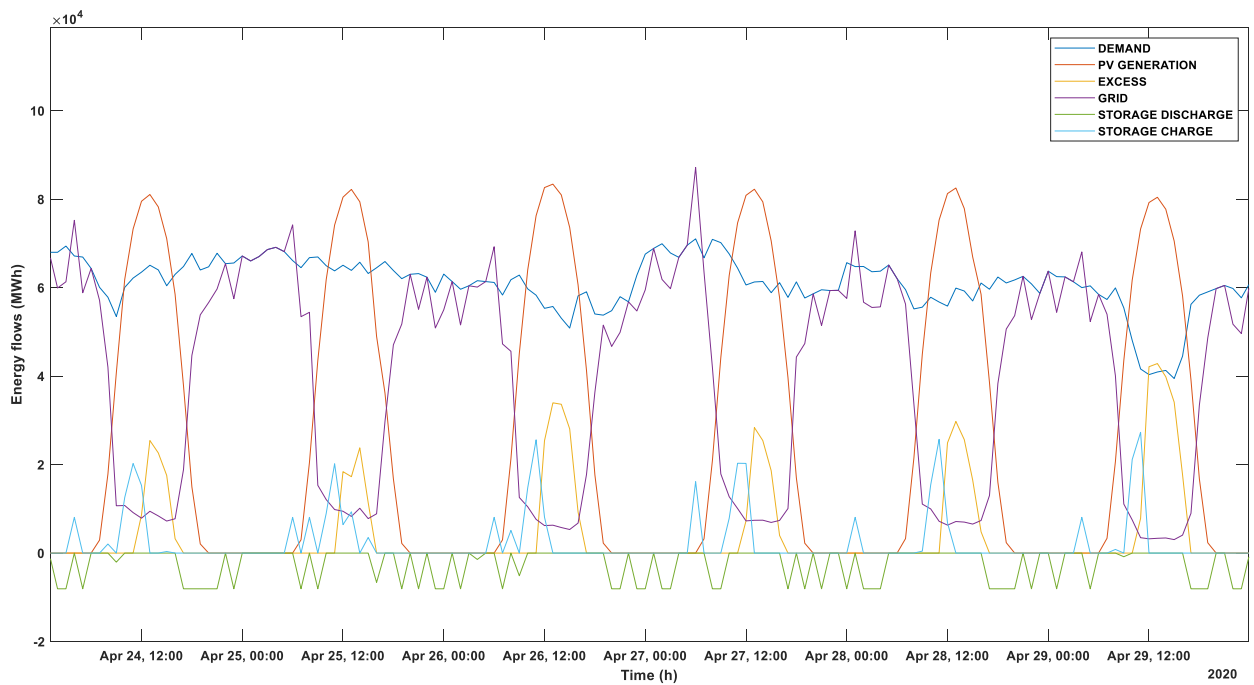


Fig. 14. Energy flows (Battery Capex Present value-25 %, RS 31.18 %).

5.3. Scenario 3

Scenario 3 involves the evaluation of four sensitivity experiments to track the influence of the financial element, the WACC. Also, the system's renewable share is also varied between 61.18 % and 31.18 %. By conducting these experiments, we aim to assess the impact of changing the WACC and renewable share on the system performance and financial outcomes. This analysis helps to understand how these parameters affect the system viability and cost effectiveness.

Table 6
Techno-economic assessment for scenario 3.

Wacc	7 %						
RS	Autonomy Factor (%)	LCOE (\$/kWh)	NPV (M\$)	PV Capacity (kWp)	Storage Capacity (kWh)	Supply Reliability (%Demand)	First investment (M\$)
61.18 %	61.18	0.11847	428.730	199.153	659.146	1	283.282
31.18 %	31.18	0.07111	257.342	100.246	68.353	1	92.838
Wacc	5 %						
61.18 %	61.18	0.10844	431.424	201.752	654.096	1	284.057
31.18 %	31.18	0.06797	270.409	101.337	66.265	1	93.164

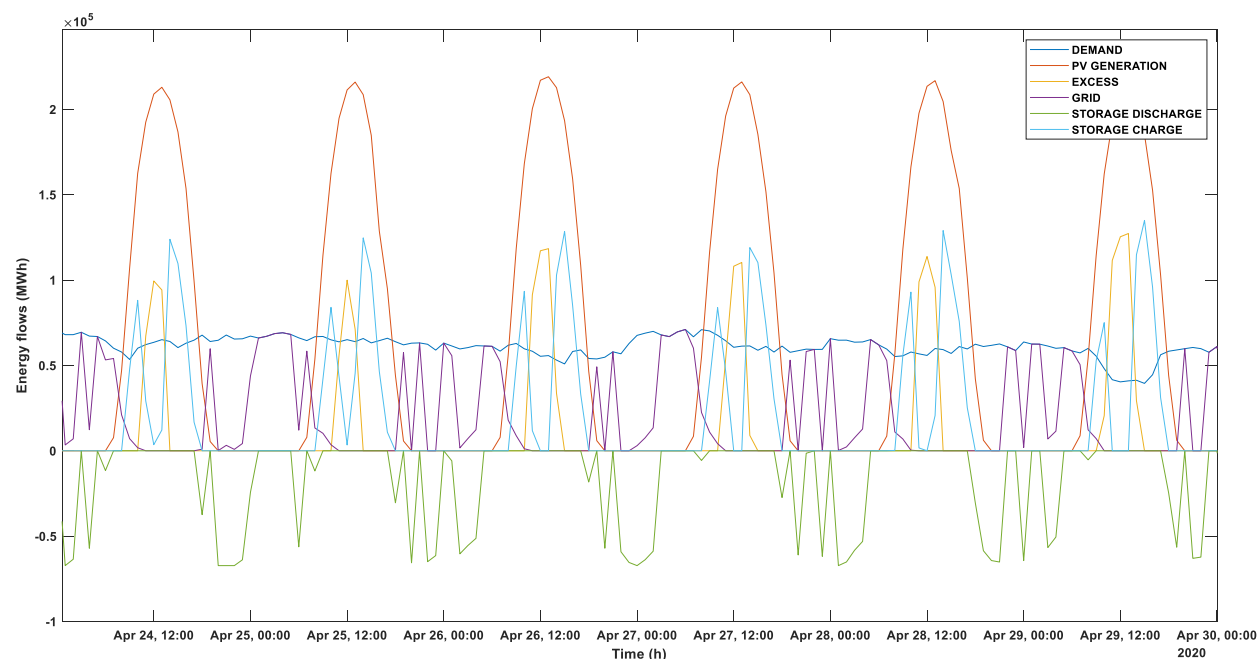


Fig. 15. Energy flows (Wacc 5 %, RS 61.18 %).

The simulation options undergo a thorough evaluation, and Table 6 presents in detail the results. These results provide insights into the optimal energy system capacities and costs that meet the total demand while adhering to the specified constraints. By examining these options, we can identify the most efficient and cost-effective configuration for the energy system.

Figs. 15–18 provide a comprehensive illustration of the energy flows within the energy system, showcasing the interactions between production, storage, and consumption components at the site. For this scenario, the optimal capacity of the PV system is 201.752 MWp for a renewable supply of 61.18 % and 101.337 MWp for 31.18 % renewable supply. These optimal capacities depend both on the value of the WACC and the site-specific load profile, characterized by a peak demand of 83.138 MW. By considering these parameters, we can effectively assess the optimal PV system capacity required to meet the specified renewable supply targets and adapt to the unique load profile of the site.

5.4. Discussion

LCOE is one of the indicators that allow the comparison of different generation technologies that differ in capital costs, operation and maintenance costs, useful life, and many other elements. It is an indicator of the price of electricity required for a project with revenues equal to costs, including the achievement of a return on investment equal to the discount rate.

In this study section, we have plotted the variation of this indicator to track the PV cost, Battery cost, and Wacc influence on this parameter and thus on the project profitability. Fig. 19 shows a decreasing value with decreasing PV investment cost itself, for example, and at a 61.18 % renewable supply, for the currently adopted PV investment cost which is about 691 \$/kWp according to the latest cost updates, the LCOE is about 0.11847 \$/kWh and it can reach 0.11096 \$/kWh if the PV investment cost decreases by 25 %. This downward trend holds for all combinations studied in the scenario 1.

The sensitivity study of the combination (battery investment cost, share of renewables) gives a variation of the financial indicator LCOE as presented in Fig. 20, it follows the behavior of the sensitivity parameters, e.g., the LCOE decreases by reducing the battery investment cost by 25 % of its current value, for a 61 % renewable supply, the LCOE goes from about 0.11847 \$/kWh to 0.10536

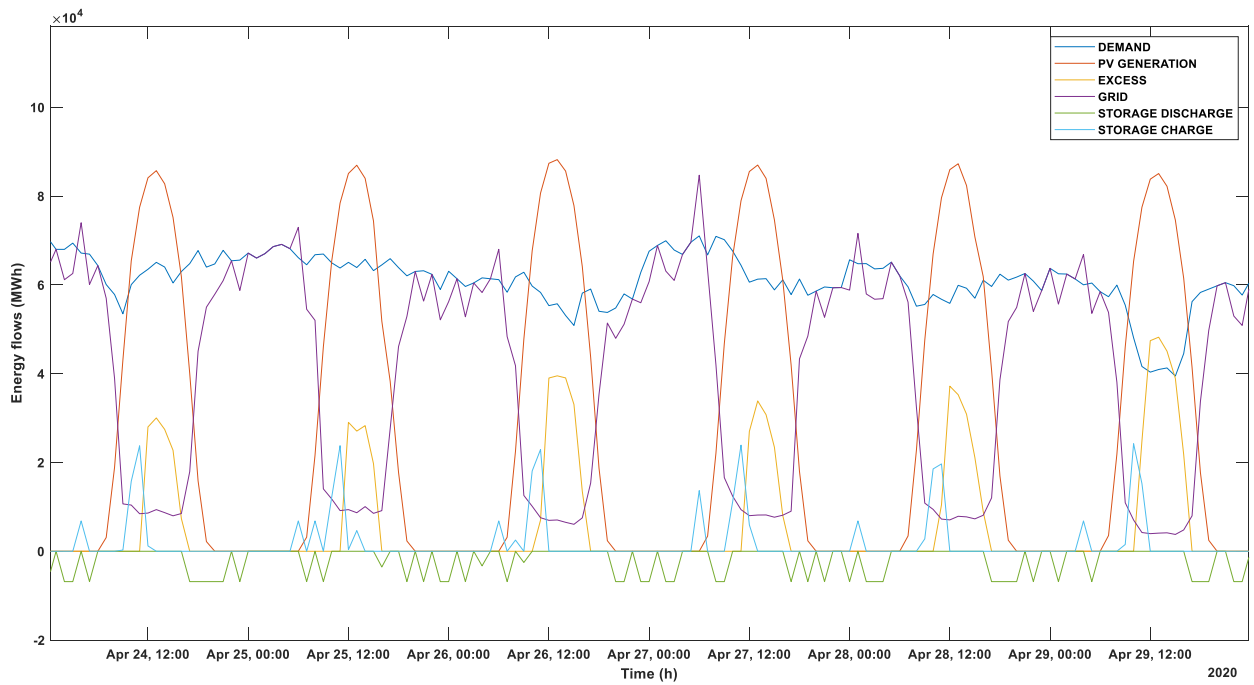


Fig. 16. Energy flows (Wacc 5 %, RS 31.18 %).

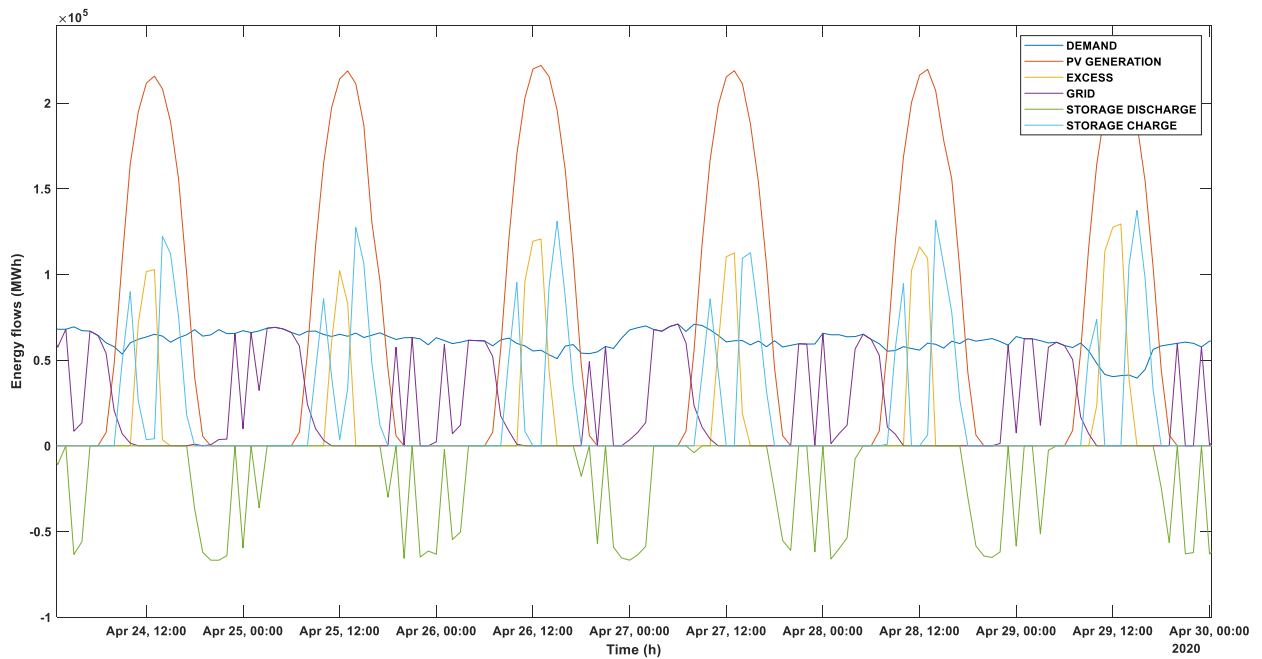


Fig. 17. Energy flows (Wacc 7 %, RS 61.18 %).

\$/kWh. This downward trend is also true for all combinations studied in the second scenario.

The combination (WACC, share of renewables) sensitivity study gives a variation of the financial indicator LCOE as presented in Fig. 21, it follows the behavior of the sensitivity parameters and the LCOE increases by increasing project wacc.

Currently, the average value adopted by energy projects in Morocco is around 7 %, which is the case giving an LCOE of about 0.11847 \$/kWh for a 61.18 % renewable supply, this value decreases to 0.10844 \$/kWh if we adopt a WACC of 5 %. So, if we decrease the WACC by 28.57 % the LCOE decreases by 8.47 % which shows that this parameter is also important, this behavior is verified for all

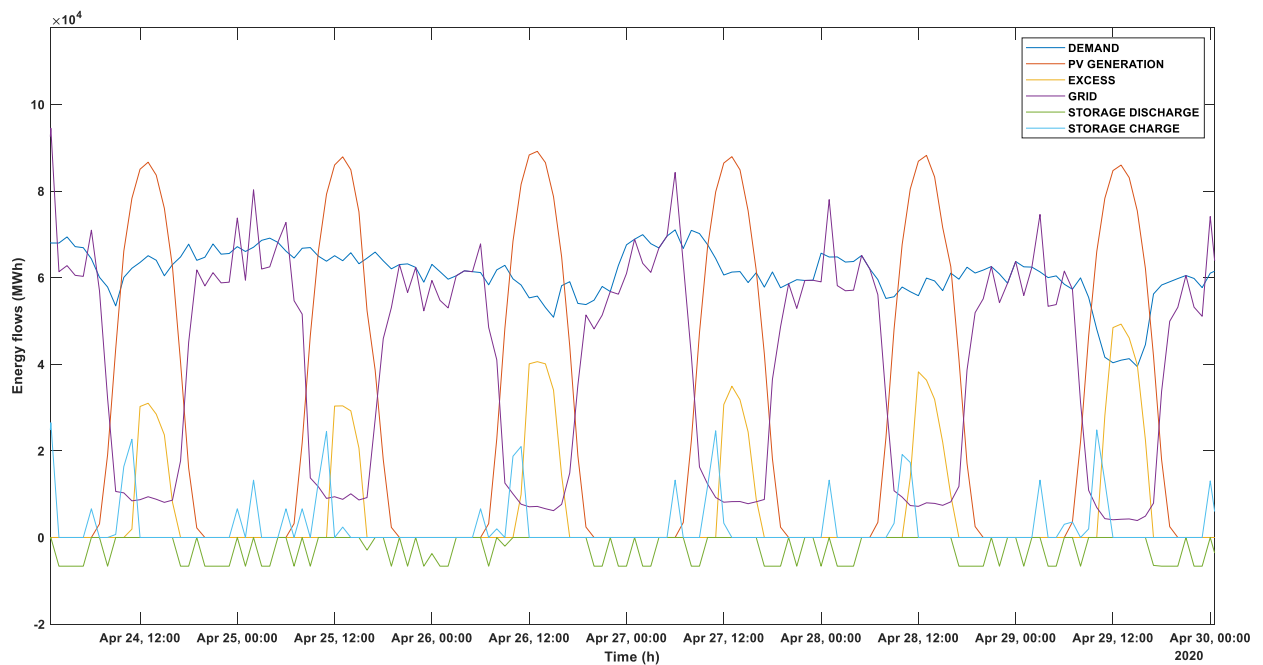


Fig. 18. Energy flows (Wacc 7 %, RS 31.18 %).

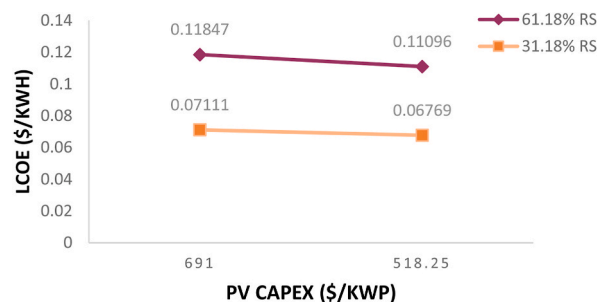


Fig. 19. LCOE variation by PV Capex and RS sensitivity.

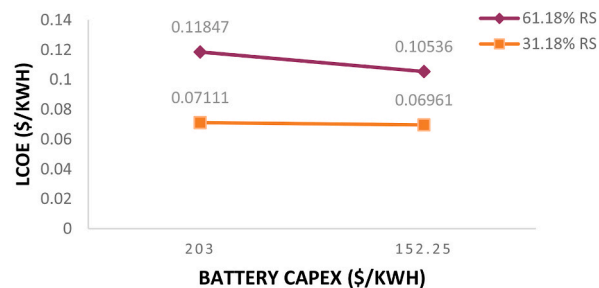


Fig. 20. LCOE variation by Battery Capex and RS sensitivity.

combinations studied in scenario 3.

Based on these financial metrics and in terms of LCOE, installation, operation, and maintenance expenses, the reference model with a PV capital cost of about \$691/kWp and a battery capital cost of about \$203/kWh and a WACC of 7 % yielded an LCOE of 0.11847 \$/kWh, the energy cost can be improved shortly due to the decreasing cost trend according to the analyses made above and that will be an increasingly cheaper solution to provide electricity, ensuring a constant supply to meet the total expected demand. A summary of the battery and PV capital cost sensitivity study showing the co-behavior of LCOE for different scenarios is illustrated in Figs. 22 and

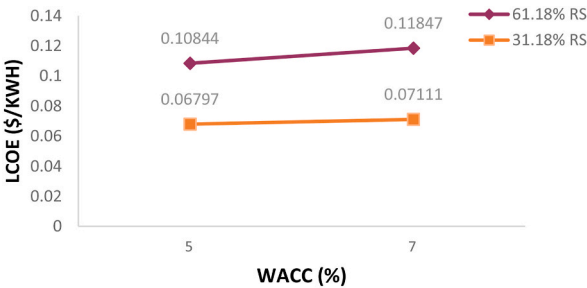


Fig. 21. LCOE variation by WACC and RS sensitivity.

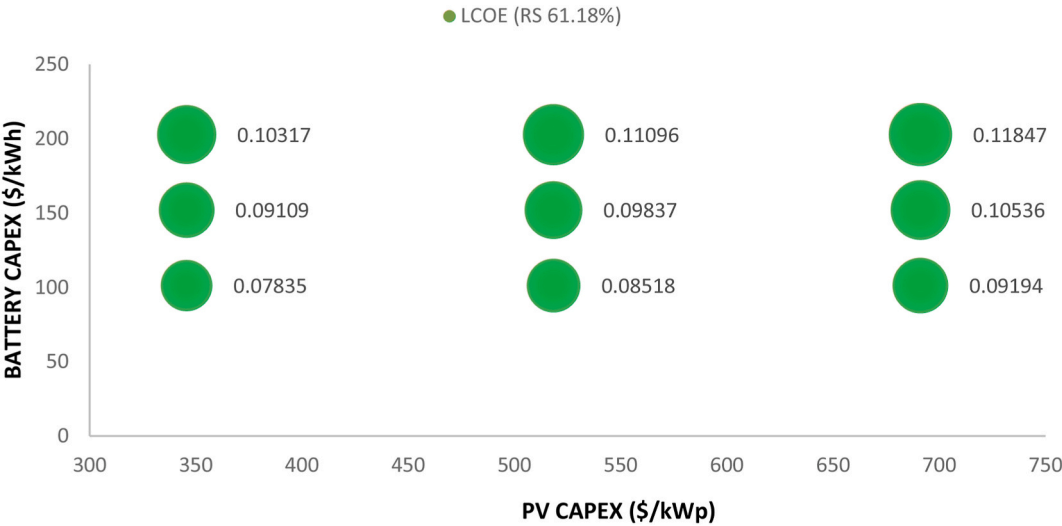


Fig. 22. LCOE variation by PV Capex, Battery Capex (RS 61.18 %).

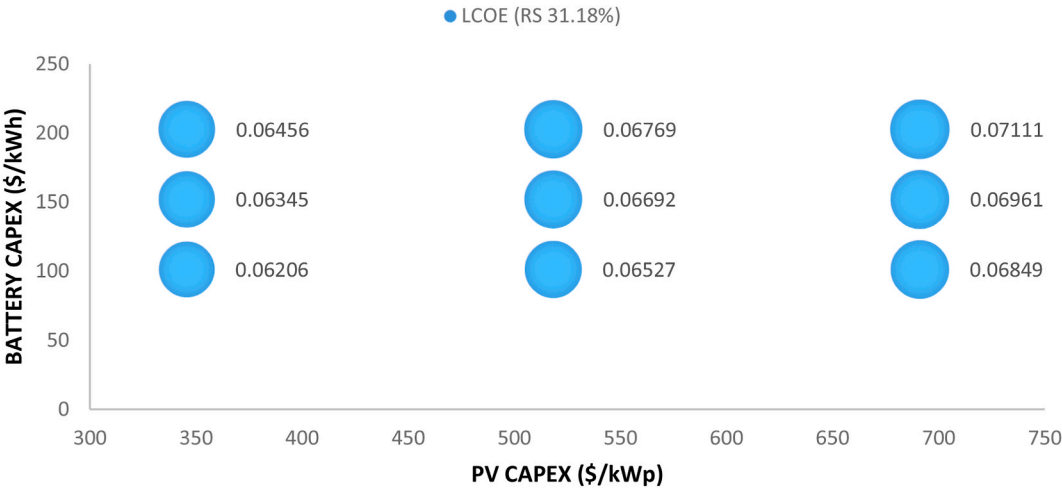


Fig. 23. LCOE variation by PV Capex, Battery Capex (RS 31.18 %).

23. The LCOE tends to decrease to competitive costs (\$0.07835/kWh) for 61.18 % renewables share and (\$0.06206/kWh) for 31.13 % renewable share when both PV and battery capital costs decrease by 50 % of their current values.

6. Conclusion

This work enabled an economic analysis of the energy system allowing the supply of electrical energy to the studied sites. A sensitivity analysis of a set of economic and financial parameters was also implemented (PV Cost investment, Battery Cost investment, WACC). The sensitivity inputs were presented under a matrix form, which allowed the sensitivity study of two parameters simultaneously generating a multitude of combinations to be simulated, discussed and evaluated in the case study. Two supply options with 61.18 % and 31.18 % renewable energy coverage are presented.

The simulation results are compared economically through the LCOE indicator, for 61.18 % renewable energy coverage, scenario 1 showed that the investment cost of PV improves the economic profitability by reducing the LCOE by 6.34 %, if decreased by 25 %. While scenario 2 showed that the investment cost of the batteries can more strongly improve this profitability if it also decreases by 25 % through a 11.06 % reduction in LCOE. From a financial point of view, the wacc parameter variation also increases its influence on the economic profitability by decreasing LCOE by 8.46 % if we have a wacc of about 5 % instead of the 7 % currently adopted. Although for a 31.18 % renewable energy system, for a 25 % reduction in PV CAPEX, the LCOE decreases by 4.8 % while it decreases by 2.11 % by reducing the battery CAPEX by 25 %. For the same share of renewable energy, if it reaches 5 %, the WACC lowers the LCOE by 4.41 %. Therefore, when tracking the evolution of LCOE for each scenario studied, we infer that the variation in battery CAPEX and WACC affects power generation costs more strongly than the variation in solar CAPEX.

Finally, we can state that the answer to the research question outlined in the case study chapter has been addressed. Based on these financial metrics and in terms of LCOE, installation, operation and maintenance expenses, the reference model with a PV capital cost of about 691 \$/kWp and a battery capital cost of about 203 \$/kWh and a WACC of 7 % yielded a LCOE of 0.11847 \$/kWh, the energy cost can be improved in the near future due to the decreasing cost trend according to the analyses made above and that will be an increasingly cheaper solution to provide electricity, ensuring a constant supply to meet the total expected demand.

Future work will investigate how degradation of battery energy storage systems affects both the cost and efficiency of their use in microgrid applications. This analysis will take into account the constraints associated with the technical specifications of BESS and the particular conditions of the grid. By better understanding how BESS degradation affects these elements, we will be able to develop more resilient strategies for their optimal use in real-world environments.

CRediT authorship contribution statement

Oumaima Mahir: Writing – original draft, Software, Methodology, Conceptualization. **Abdelilah Rochd:** Writing – review & editing, Methodology, Conceptualization. **Aboubakr Benazzouz:** Validation, Supervision, Conceptualization. **Hicham Ghennioui:** Supervision.

Data availability

The data that has been used is confidential.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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